

NORDIC BIOBANK × NVIDIA HACKATHON · 2026

Everything OMOP

Federated learning on harmonised biobank data

SUBPROJECT 1

Automated ETL

Raw biobank data → OMOP conversion and quality control

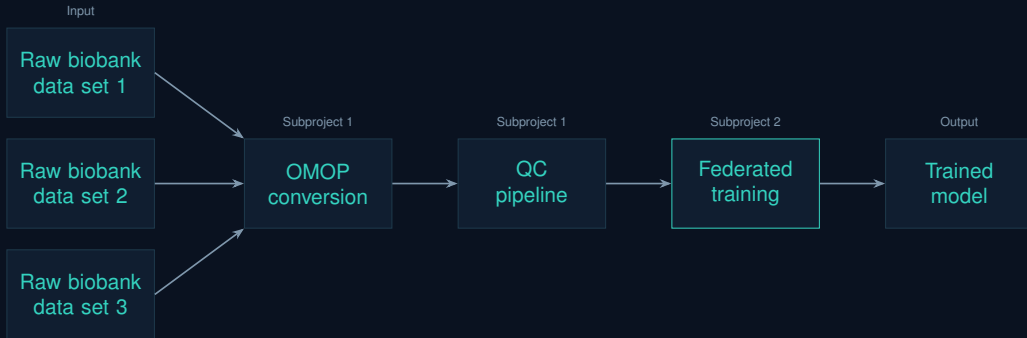
SUBPROJECT 2

Federated learning

Proof of concept on OMOP datasets

OVERVIEW

End-to-end pipeline



Every box exists. They are connected.

SUBPROJECT 1

OMOP conversion & quality control

THE PROBLEM

- Every biobank stores its data differently
- OMOP makes analysis portable across sites
- Status quo: hand-built ETL per source, months of work

OUR APPROACH

- An agent skill that converts
- Not a script written for one dataset
- Scope: minimal table set, synthetic data, measurable

SUBPROJECT 1

OMOP conversion & quality control

CONVERSION PIPELINE

- Profile the source: fields, codes, distributions
- Propose structural mapping: source field → OMOP field
- Propose vocabulary mapping: source code → standard concept
- **Clinical sign-off** before a mapping is accepted
- Run ETL: contracted tables, one directory per site

QUALITY CONTROL

- Reference: published expert mappings (EHDEN, The Hyve)
- Expert mappings recovered, correctly and wrongly
- Share of records on `concept_id = 0`
- Ten most frequent unmapped codes
- Output handed over via a fixed data contract

SUBPROJECT 2

Federated learning on OMOP data

IMPLEMENTATION

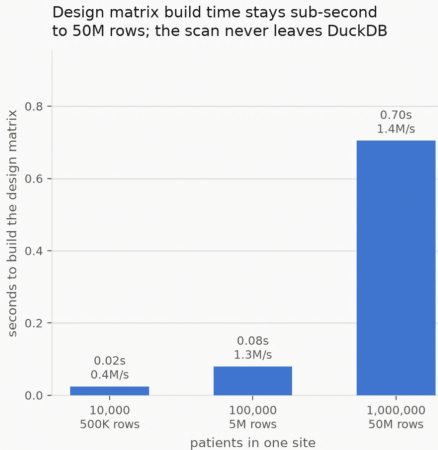
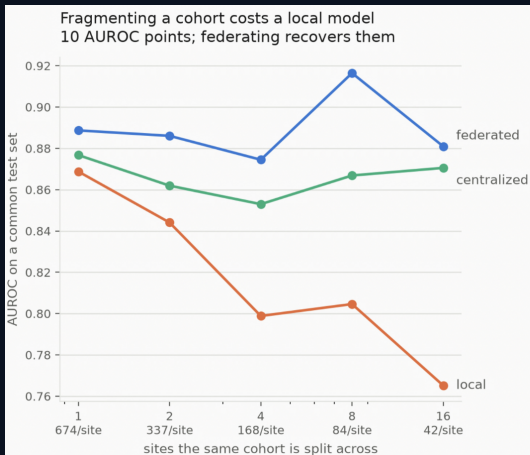
- DuckDB enables lazy loading
- Time series stored as 3D sparse arrays: patients $\times$ concepts $\times$ time
- Custom PyTorch dataloader, validated on 5 Synthea OMOP cohorts
- `omopflare` exposes one API, integrates with NVFlare

SCOPE & OBJECTIVE

- Cox proportional hazards across simulated federated sites
- Goal is end-to-end pipeline functionality, not predictive performance
- Consumes OMOP output from subproject 1 directly

SUBPROJECT 2

Why federate, and what it costs



Federated learning on OMOP data

BIOMARKERS

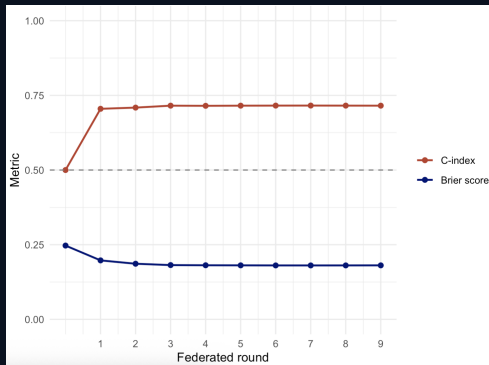
BMI
SBP
HbA1c
Glucose
Creatinine
eGFR

MODEL

- Cox proportional hazards, 5-year survival
- Predictors: age and blood-based biomarkers
- Federation framework in place, single cohort for the first demo
- Nine federated rounds: C-index ≈ 0.71 , Brier score stable

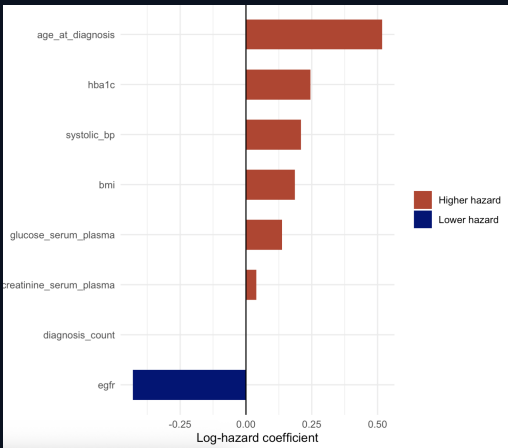
SUBPROJECT 2 · PROOF OF CONCEPT ON SIMULATED DATA

Nine federated rounds



SUBPROJECT 2 · PROOF OF CONCEPT ON SIMULATED DATA

The coefficients point the right way



- Age, HbA1c, systolic pressure and BMI raise the hazard
- eGFR lowers it
- On synthetic data this is not biology
- It shows the pipeline is wired correctly, end to end

WHAT WE ARE CLAIMING

A complete path from raw biobank data to a federated model.

The data never left its site.

Not predictive performance. Synthetic data throughout. Measurable at every step.

github.com/collaborativebioinformatics/Everything_OMOP